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Maximum Likelihood Reinforcement Learning

REINFORCE Optimizes a Lower-Bound Approximation of True Likelihood — MaxRL Fixes This with More Sampling Compute

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Standard reinforcement learning for LLMs does not maximize the likelihood of correct rollouts — it optimizes a Jensen lower bound that allows probability to diffuse across mediocre samples. *Maximum Likelihood Reinforcement Learning* (MaxRL) introduces a compute-indexed family of objectives that converge to exact maximum likelihood as more rollout samples are drawn per query, achieving consistent improvements on math reasoning, navigation, and large-scale Qwen3 training at no increase in gradient steps.

AT A GLANCE

Problem: REINFORCE optimizes only a lower-order approximation of true maximum likelihood over correct rollouts in sampling-based RL settings.

Why it matters: The objective gap causes probability mass to diffuse across mediocre rollouts rather than concentrating on correct answers.

Method: Importance-weighted MLE estimator with k samples per query, interpolating between REINFORCE ($k = 1$) and exact MLE ($k \to \infty$).

Result: Consistent improvement on maze navigation, GSM8K, and large-scale Qwen3 training; gains scale with k.

Evidence: Toy classification verifies closeness to MLE; ablations confirm the importance-weighting is the key ingredient.

The Big Picture

Training language models with reinforcement learning on binary correctness signals — did the generated code pass the tests? does the math proof check out? — is the dominant paradigm for post-training. The underlying objective is maximum likelihood: given that a correct response exists, maximize the probability the model assigns to it.

But the paper reveals a gap. REINFORCE, the workhorse of this paradigm, does not maximize this likelihood. It optimizes a lower bound, one that is tight only when the model already concentrates most probability on correct responses. In practice, especially early in training, REINFORCE wastes gradient signal on rolling updates that nudge mediocre rollouts upward rather than concentrating on correct ones.

MaxRL closes this gap by drawing $k > 1$ samples per query and constructing an importance-weighted objective whose gradient converges to the true MLE gradient as k grows. More sampling compute buys a closer approximation to the ideal training signal.

Why Existing Approaches Fall Short

REINFORCE: With one sample per query and binary reward, the REINFORCE gradient is:

$$\nabla_\theta J_{\text{REINFORCE}} = \mathbb{E}_{y\sim\pi_\theta}[r(y)\nabla_\theta \log \pi_\theta(y|x)]\,.$$

This equals the gradient of $\mathbb{E}_x[\mathbb{E}_{y\sim\pi_\theta}[r(y)\log \pi_\theta(y|x)]]$, a lower bound on $\mathbb{E}_x[\log \sum_{y\in Y^+} \pi_\theta(y|x)]$ by Jensen's inequality. The gap is non-negligible when π_θ spreads probability across many incorrect responses.

Group Relative Policy Optimization (GRPO): Draws k samples but uses them for reward normalization rather than MLE approximation; the resulting objective is still a REINFORCE variant, not a MLE estimator.

Core Mechanism

MaxRL forms an importance-weighted estimate of the true MLE gradient using k rollouts per query. The weights rescale each rollout's contribution proportionally to the model's current assignment of probability to that rollout among all correct rollouts sampled:

$$w_i = \frac{r(y_i)\,\pi_\theta(y_i|x)}{\displaystyle\sum_{j=1}^{k} r(y_j)\,\pi_\theta(y_j|x)}.$$

The MaxRL objective is:

$$J^{(k)}_{\text{MaxRL}}(\theta) = \mathbb{E}_{x,\,y_{1:k}\sim\pi_\theta}\left[\log \frac{1}{k}\sum_{i=1}^{k} r(y_i)\,\pi_\theta(y_i|x)\right],$$

whose gradient is:

$$\nabla_\theta J^{(k)}_{\text{MaxRL}} = \mathbb{E}\left[\sum_{i=1}^{k} w_i\,\nabla_\theta \log \pi_\theta(y_i|x)\right].$$

At $k = 1$ this reduces to REINFORCE; as $k \to \infty$ this converges to the exact MLE gradient almost surely.

Technical Formulation

Let Y^+ be the set of correct responses for query x. The true maximum likelihood objective is:

$$J_{\mathrm{MLE}}(\theta) = \mathbb{E}_x \left[\log \sum_{y \in Y^+} \pi_\theta(y|x) \right].$$

REINFORCE lower bound: By Jensen's inequality applied to the concave log,

$$J_{\mathrm{REINFORCE}}(\theta) \leq J_{\mathrm{MLE}}(\theta),$$

with equality only when all probability mass is on a single correct response.

MaxRL convergence:

$$J^{(k)}_{\mathrm{MaxRL}}(\theta) \xrightarrow{k \to \infty} J_{\mathrm{MLE}}(\theta) \quad \text{a.s.}$$

by the law of large numbers applied to the importance-weighted estimator.

The gap between MaxRL and MLE at finite k is bounded by:

$$|J_{\mathrm{MLE}} - J^{(k)}_{\mathrm{MaxRL}}| = O\left(k^{-1/2}\right)$$

under mild regularity conditions on π_θ.

Learning or Inference Procedure

1. For each query x, sample k rollouts $y_1, \ldots, y_k \sim \pi_\theta(\cdot|x)$.
2. Evaluate reward $r(y_i) \in \{0, 1\}$ for each rollout.
3. Compute importance weights w_i using current policy probabilities.
4. Compute $\nabla_\theta J^{(k)}_{\mathrm{MaxRL}}$ using the weighted gradient estimate.
5. Update θ via Adam.
6. At inference: standard autoregressive sampling (MaxRL changes only the training objective, not the inference procedure).

What the Guarantee Says

MaxRL is a consistent estimator of the MLE gradient: as k grows, the training objective converges to exact maximum likelihood. The rate of convergence is $O(k^{-1/2})$ in the objective value. No distributional shift correction or importance weight clipping is required as long as π_θ remains supported on Y^+.

Experimental Findings

- **Toy classification:** MaxRL closely tracks exact MLE training signal; REINFORCE lags by a measurable margin throughout.
- **Maze navigation:** MaxRL achieves higher success rate than REINFORCE and GRPO at equivalent gradient steps.
- **GSM8K:** 3–5% accuracy improvement over REINFORCE with the same compute; gains scale with k up to $k = 16$.
- **Qwen3 (large-scale):** Consistent accuracy improvement; MaxRL with $k = 8$ matches REINFORCE with $k = 1$ at roughly half the gradient steps.

Ablations and Interpretation

Effect of k: Performance improves monotonically with k up to $k \approx 16$, beyond which gains plateau (model capacity becomes the bottleneck, not objective quality).

Importance weights vs. uniform: Replacing w_i with uniform weights (equivalent to supervised fine-tuning on correct samples) degrades performance, confirming that the reweighting term is the key ingredient, not merely drawing more samples.

Relationship to GRPO: GRPO's reward normalization does not approximate MLE; it is a variance-reduction technique within the REINFORCE objective. MaxRL's improvement is orthogonal and additive to GRPO's tricks.

Reference

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